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A.I. LIVSHITS✉ ,*
V.N. ALIMOV*
M.E. NOTKIN*
M. BACAL

Hydrogen superpermeation resistant to ion sputtering

Laboratoire de Physique et Technologie des Plasmas V.M.R. 7648 du C.N.R.S., École Polytechnique, 91128 Palaiseau, France

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ABSTRACT A nonmetallic monolayer on the surface of a metallic membrane gives rise to the superpermeation of suprathermal hydrogen. This means that virtually the whole implanted flux passes through the membrane, depending neither on membrane temperature nor on its thickness. The fact that superpermeation is degrading at a sufficiently high energy of the incident hydrogen particles was considered to be a 'natural' result of sputtering of the monolayer. This view was equally applied to Nb, one of the best metals for superpermeation. It is shown in this article that Nb containing O impurity will be normally superpermeable to suprathermal hydrogen at any incident energy, the earlier observed sensitivity to sputtering being conditioned by the NbC layer originally present at the 'real' Nb surface. The dynamic exchange between the surface and bulk O impurity is proven to be responsible for the superpermeability of Nb resistant to sputtering. When the NbC layer is present, it hinders such an exchange.

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1 Introduction

A metal surface in a vacuum is typically covered by a nonmetal monolayer (the 'real' surface). For example, if a metal is heated in a vacuum, virgin polyatomic films are desorbed or dissolved, with only a monolayer of nonmetallic impurity remaining at the surface. Such a monolayer gives rise to a potential barrier impeding both the dissociative absorption and associative desorption of hydrogen in the form of thermal H_2 molecules [1]. If, however, the metal is acted upon by suprathermal hydrogen particles whose energy (kinetic, internal or chemical) exceeds ~ 1 eV, the barrier does not prevent their absorption, while continuing to suppress the release of the hydrogen thermalized in the lattice. That increases hydrogen retention in, and permeation through, the metal by many orders of magnitude [1]. Reemission may be impeded to such an extent that implanted H atoms cross the plane membrane sample many times before the release. If the two membrane surfaces are identical (which is typical for 'natural' surfaces), a half of the implanted flux passes through the membrane, irrespective of its temperature and thickness. Thus the membrane becomes 'superpermeable', i.e., almost 'transparent' to suprathermal hydrogen [1]. The membranes superpermeable to energetic hydrogen are at the same time virtually impermeable to thermal hydrogen molecules and to other gases. Because of this, the superpermeable membranes are able to automatically compress by orders of magnitude the permeating hydrogen, and to purify it from any impurities, including He [1]. These features allow applications of superpermeable membranes for hydrogen handling, particularly, in nuclear fusion devices for fuel (D/T) handling and particle flux control [1–3].

Given the above role of a nonmetallic monolayer, one naturally would expect sputtering to produce a significant effect. Indeed, the permeation spike is typically observed: the permeation flux first rises upon the initiation of a flux of hydrogen ions of an energy, E_{inc}, on keV scale incident on a membrane of V [4], Ni, Mo [5], Fe [6] or stainless steel [1, 7], but then passes through a maximum to eventually drop by orders of magnitude [1].

The energy dependence of the effects of ion bombardment on hydrogen permeation was systematically investigated for a membrane of polycrystalline Nb [8–10]. Niobium was chosen because it is, along with other Group Va metals, one of the best materials for superpermeation [1]. In the case of Nb, the nonmetallic monolayer would typically be formed by O, usually, with a NbO stoichiometry [11], which remains after the dissolution of a virgin oxide film at metal heating [12] or segregates from the bulk at elevated temperatures, if the surface has been preliminary cleaned [11, 13]. When E_{inc} exceeds the threshold of O monolayer sputtering ($\sim$50 eV), the initial plasma driven superpermeation degrades, with an orders-of-magnitude permeation drop occurring. Such a result was obtained at different laboratories on Nb samples of Russian and Japanese make, both in ion beam [1, 14] and plasma [8–10] experiments. Since the stable operation under ion bombardment is a key point for applications of superpermeable membranes in the nuclear fusion [2, 3], special efforts were made to enhance membrane resistance to sputtering [9, 10]. A definite progress was made, but the effect of sputtering still remained rather strong.

This work prompted by the observation of an unexpected phenomenon [15]: the effects of sputtering disappeared entirely after an in situ heating of Nb membrane at $T \sim 1450\,^{\circ}$C

✉ Fax: +7-812/587-9334, E-mail:livshits@qrel.sut.ru
Permanent address: Bonch-Bruyevich University, 61 Moika, St. Petersburg 191186, Russia

(it was only heated to $T \leq 1280\,^{\circ}\mathrm{C}$ before). The superpermeation was keeping stable over the whole energy range investigated (up to 1 keV) at a flux density of incident ions as high as a few $\mathrm{mA/cm^2}$.

One can expect other related phenomena associated with the hydrogen dissociation barrier (such as an enhanced absorption and the suppressed reemission [1]) to become insensitive to sputtering as well alongside of hydrogen permeation. Hence the phenomena taking place at metal interactions with energetic hydrogen (tritium, in fusion devices) may be radically dependent on whether the plasma facing material is in a sputter-sensitive or in a sputter-resistant state. On account of the practical importance of the issue, the purpose of this work was to investigate the observed phenomenon and to elucidate its physical mechanisms.

After a brief presentation of the experimental techniques employed, the first observation of the sputter-resistant superpermeation is described, and results of a more systematic experimental investigation are presented; then, the underlying physico–chemical origin of the phenomenon is suggested complemented by an analysis of the mechanisms involved.

2 Experimental

The experiments described below were staged with the same plasma-membrane apparatus where the effects of ion bombardment on the superpermeability of Nb were investigated previously [8, 10]. A membrane sample in the form of a tube with one closed end is immersed in hydrogen plasma (of $\sim 10^{11}\ \mathrm{cm^{-3}}$ density, ~ 1 eV electron temperature, and $\sim 3 \times 10^{-3}$ Torr H_2 pressure) uniformly filling a 0.1 $\mathrm{m^3}$ vessel. Two turbomolecular pumps are continuously evacuating the down- and upstream sides of the membrane sample during the plasma operation. Partial gas pressures are gauged with a differentially pumped quadrupole gas analyzer. The gas impurity pressure in the plasma was $\sim 1 \times 10^{-6}$ Torr (mainly CO).

Incident ion energy, E_{inc}, was varied by changing the negative membrane bias with respect to plasma (in fact, with respect to the chamber wall serving as the discharge anode [8, 10]). The ion flux (mainly H_2^+ and H_3^+) incident on the membrane sample is typically $\sim 2\,\mathrm{mA/cm^2}$ and is kept approximately constant at bias variations [8].

Tubular membrane samples (1 cm diameter, 18 cm length, and 0.01 cm wall thickness) were made of 99.95-purity polycrystalline Nb. The sample temperature was regulated by ohmic heating and could be varied from 20 °C up to the material melting point. The temperature was measured with a standard WRe (5%)–WRe(20%) thermocouple located inside the tubular membrane.

The membrane sample was heated in a vacuum up to 1280 °C before the commencement of experimental campaign. Oxygen from virgin niobium oxide surface films is getting dissolved and uniformly distributed over the membrane bulk even at substantially lower T [16], with only an O monolayer remaining at either membrane surface [8, 11–13].

A layer-by-layer surface analysis of the Nb samples before and after operation is accomplished by means of Auger Electron Spectroscopy (AES) in a separate device.

3 Experimental results

3.1 *Plasma driven permeation in the 'usual' state*

In the previous investigations, when the membranes were only heated to temperatures not exceeding 1280 °C, the following pattern of behavior was typically observed [8–10]. The membrane samples exhibited plasma driven superpermeation over the incident energy range where there is no sputtering (e.g., at membrane floating potential). An evidences of superpermeation is the temperature independence of the permeation flux (Fig. 1, curve 1). Superpermeation existed in a finite range of low E_{inc} (e.g., at $E_{\mathrm{inc}} < 50$ eV, Fig. 2, curve 1), only interrupted by a relatively shallow minimum at $E_{\mathrm{inc}} \approx 10$ eV which originates, most probably, from the chemical sputtering of O monolayer [8, 9]. However, superpermeation degraded at higher energies (above 50–100 eV, Fig. 2 curves 1–3) due to the physical sputtering of O monolayer: each next energy increase reduced the permeation down to a next, lower steady

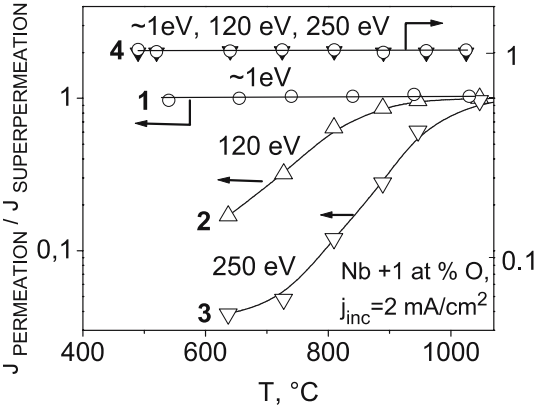


FIGURE 1 Temperature dependence of the flux of plasma driven permeation (normalized to the flux at superpermeation) through the same Nb membrane in the states sensitive (1–3, *left axis*) and resistant (4, *right axis*) to sputtering for three values of incident ion energy: ~ 1 eV (floating potential), 120 eV and 250 eV

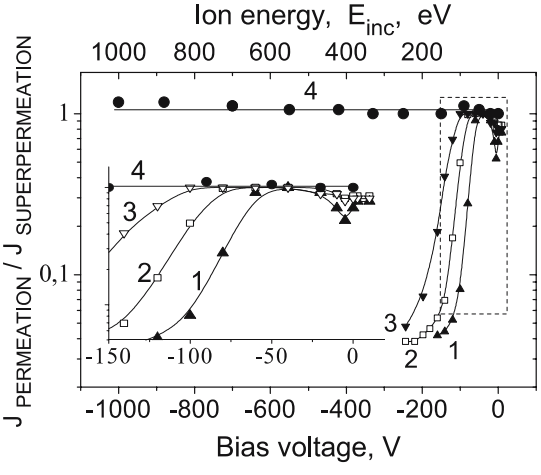


FIGURE 2 Dependence on incident ion energy (bias) of plasma driven permeation through Nb membrane under similar conditions ($T = 750\,^{\circ}\mathrm{C}$, ion flux density $\sim 2\,\mathrm{mA/cm^2}$): 1–3 - a state with the usual sensitivity to sputtering at different O concentrations in the lattice: (1) natural O impurity, (2) and (3) 1 at. % and 2.5 at. % of extra O, respectively; (4) the state resistant to sputtering

state level, with a final orders-of-magnitude decrease at $E_{inc} = 100–200$ eV. Correspondingly, the permeation flux was becoming temperature dependent (Fig. 1, curves 2 and 3). With ion energy decreased back, the permeation flux goes back up to the initial superpermeation level.

This 'spontaneous' superpermeation recovery was supposed [8–10] to be due to the restoration of O monolayer by surface segregation of O dissolved in a metal lattice [11–13]. Basing on this hypothesis, attempts were made to accelerate the surface segregation of O in order to decrease the effects of sputtering. For this purpose, the membrane was doped with an extra oxygen [9, 10]. This reduced the effects of sputtering, but the effects still remained quite substantial (cf. curves 1-3 in Fig. 2). Increasing T seemed to be another obvious way to enhance the segregation of O (Fig. 1, curves 2 and 3): as a matter of fact, the effects of sputtering were decreasing with increasing T; the superpermeation even might remain stable under sputtering, but the required temperatures were very high (> 1000 °C).

3.2 *Observation of a superpermeation stable under sputtering*

After all these attempts aimed at reducing the effect of sputtering, one was really surprised to see this effect completely disappear after the membrane in situ heating for a few minutes at 1450 °C (T did never exceed 1280 °C before). It turned out that a negative bias in our usual experimental range 0–250 V applied to the membrane had no effect on permeation; then we enhanced the negative bias voltage to 1 kV, but superpermeation remained stable in this range too: see curve 4 in Fig. 2.

Note that plasma driven permeation flux does not depend on E_{inc} in the state resistant to sputtering, even though the coefficient of incident ion implantation does. This occurs because H atoms of constant energy comprise the major part of the flux of suprathermal hydrogen from the plasma incident on the membrane [8, 17], while ions of variable energy may change the membrane permeability to suprathermal particles (including H atoms) through affecting the nonmetallic monolayer (as that actually occurs in the case of sputter resistant state).

The superpermeation was found to be resistant to sputtering over the whole temperature range investigated (Fig. 1, curve 4), including the lowermost temperatures (~ 500 °C) that we could maintain with the radiation cooling of membrane specimens at our regular ion flux (~ 2 mA/cm^2). Correspondingly, permeation was found to be temperature independent under sputtering, in contrast to that in the usual case (compare curves 2 and 3 with curve 4 in Fig. 1).

3.3 *Effects of subsurface carbide layer*

To make a hypothesis as of the underlying cause of the effect of heating, we assumed that both the upstream and downstream surfaces of the Nb membrane most probably had had monolayers of O not immediately on the metal, but on a polyatomic subsurface carbide layer of typically present at the carbide forming metals. Heating to $T > 1450$ °C removes the carbide. Our idea was that the emergence of sputter-resistant superpermeation had its origin in this sample modification.

Such a hypothesis had a base in the following reasons: (1) ~ 1400 °C is the critical temperature at which the decomposition of NbC into Nb and CO in a reaction with solute O is starting [16], and, the release of CO in an amount as large as $\sim 10^{18}$ cm^{-2} was actually observed at sample heating in the range 1450–1550 °C. At approximately the same temperature, the limit of C solubility in Nb reaches a level of 0.1 at. % [16], and hence subsurface carbide layers of a thickness on the order of 0.1 μm can be completely dissolved in the 100 μm-thick membrane sample. (2) A layer-by-layer AES analysis shows that the original surface of manufactured Nb foils is typically covered with a polyatomic carbide layer (Fig. 3). AES analysis of the samples subjected to a long-term exposition in our plasma has shown that, in contrast to an easy dissolution of the virgin oxides, the polyatomic carbide layers stay at the both sides of the membrane sample at least for hundreds of hours, even despite significant heating (1280 °C). (3) Hydrogen transport through even rather thick (e.g., ~ 1 μm) NbC/Nb$_2$C layers has been recently shown [3] to occur quickly enough so as not to affect Nb superpermeability.

To directly examine the effects of the carbide layer, a polyatomic C layer was deposited in situ, from plasma in a mixture of $H_2 + 10\%CH_4$, onto the upstream surface of the decarbonized membrane resistant to sputtering. As it was found later by a controlled removal of C in the form of CO, the deposited amount was $\sim 10^{18}$ °C/cm^2. Like before [3], the deposition of polyatomic C onto a relatively cold (700 °C) Nb resulted in a drastic decrease of plasma driven permeation, but a subsequent carbidization of this layer by heating to 1200 °C completely restored superpermeability in the energy range below the sputtering threshold . However, the membrane became sensitive to sputtering again (Fig. 4), with all the typical energy dependence peculiarities observed (compare Fig. 4 to curves 1–3 in Fig. 2 [8–10]): the non-monotonic behavior around ~ 10 eV, and an orders-of-magnitude drop at $E_{inc} > 50$ eV.

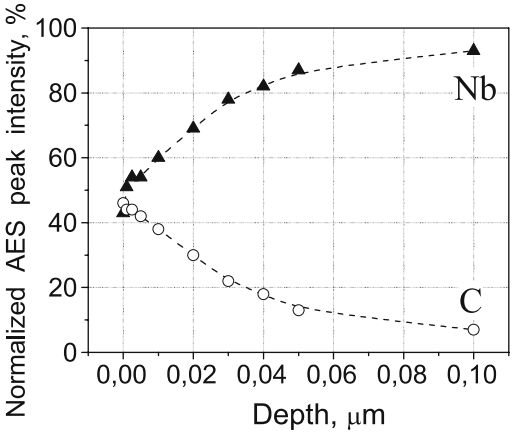


FIGURE 3 Depth profile of C distribution in Nb foil employed for the membrane material, obtained by layer-by-layer Auger Electron Spectroscopy (AES) analysis after the sample has bean heated to ~ 800 °C in the AES device. Carbon is found to be in the form of carbides (NbC/Nb$_2$C), as indicated by AES peak shapes

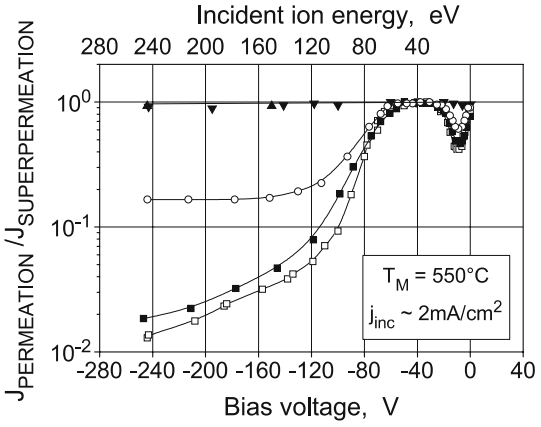


FIGURE 4 Dependencies of the plasma driven permeation on incident ion energy (bias voltage) Nb membrane in different carbidization states: ▲-after the removal of natural NbC layer, □-after the formation of NbC at upstream surface ($\sim 10^3$ monolayers), ■,○,▼-upon removal of ~ 20 at. %, 50 at. %, and 100 at. % of deposited C, respectively

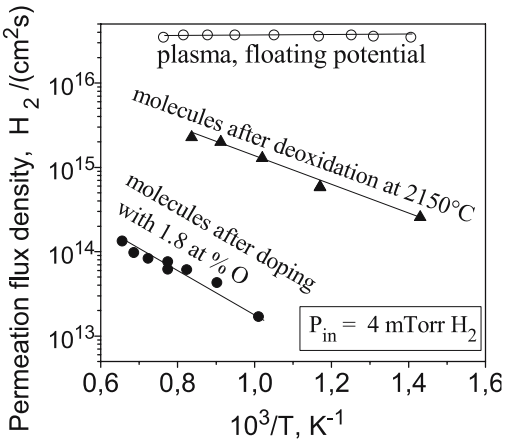


FIGURE 5 Temperature dependence of hydrogen permeation through Nb membrane: for molecules (the plasma is switched off) at a low and a high concentration of solute O, and with the plasma and membrane at a floating potential (plasma driven permeation/superpermeation is not sensitive to O concentration at the floating potential on the membrane, see Fig. 6)

Then a controlled step-by-step removal of C by its conversion into CO was carried out. Burn-out of C gradually decreased the effect of sputtering (Fig. 4), ultimately returning the membrane back into a sputter-resistant state.

Interestingly, the Nb surface noticeably darkened after the carbidization and brightened after the decarbidization, with the membrane temperature changing at the same heating current (e.g., from 1130–1280 °C) due to a change in radiation cooling (a similar brightening was observed after heating to $T > 1450$ °C of a sample having natural surface).

Thus, it has been established that underlying the 'usual' degradation of superpermeability under ion sputtering is the polyatomic carbide layer typically present at a natural Nb surface (probably, at surfaces of other carbide-forming metals as well).

3.4 *The role of solute* O *in the case of sputter-resistant state*

According to the above (Fig. 2), plasma driven permeation under sputtering significantly depends on the concentration of solute O in the case of sputter-sensitive state. In order to clarify the bearing the solute O has on conservation of the state resistant to sputtering, permeation behavior of one and the same Nb membrane containing a relatively low and a very high concentrations of O was investigated. First, the superpermeable membrane resistant to sputtering was heated up to a temperature as high as 2150 °C; Nb is getting deoxidized at this temperature through oxide evaporation [11–13, 16]. Such a treatment brought about a radical increase in the permeation of thermal H_2 molecules (see the temperature dependence presented in Fig. 5). Then the membrane was doped in situ with O to a concentration as high as 1.8 at. % (the procedure is described elsewhere [10]). The consequence was an orders-of-magnitude decrease in the molecule driven permeation: see Fig. 5.

The influence of the content of solute O on the permeation of energetic hydrogen through a sputter resistant membrane is, on the contrary, relatively weak (Fig. 6): the dependence of plasma driven permeation on incident ion energy remains much weaker than for the sputter-sensitive membrane (cf.

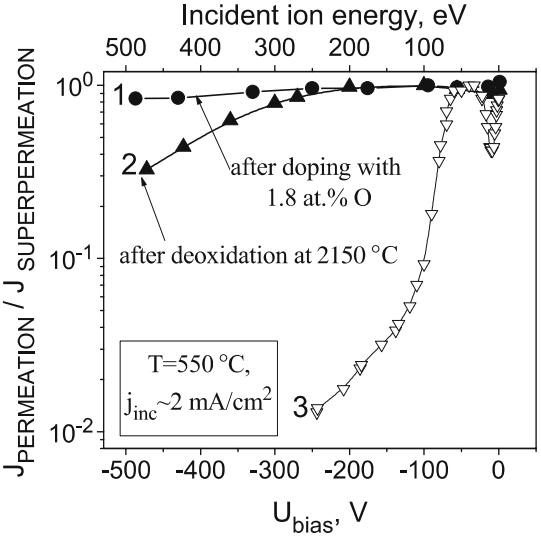


FIGURE 6 Bias/energy dependence of the plasma driven permeation through Nb membrane in the state resistant to sputtering at a high (1) and a low (2) concentration of O in the lattice (cf. to Fig. 5). The case of the 'usual' sensitivity to sputtering (3) is presented for a comparison

curves 2 and 3 in Fig. 6), even with the combination of a low O concentration and low T(~ 550 °C). However, the O concentration decrease does produce a permeation decrease with incident energy rising (cf. curves 1 and 2 in Fig. 6, see also curve 4 in Fig. 2). It should be noted that O content has no effect on plasma driven permeation (superpermeation) in both the sputter-resistant (Fig. 6) and sputter-sensitive states (Fig. 2), when there is no sputtering (e.g. at a zero bias).

3.5 *Transient processes after applying a bias*

Typical examples of the permeation decay due to sputtering are presented in Fig. 7 for both the sputter-sensitive (due to a NbC layer) and sputter-resistant (decarbidized) states. The state resistant to sputtering corresponds to a low concentration of solute O, with a permeation decrease still occurring in the range of low T and high E_{inc} (curve 2 in Fig. 6).

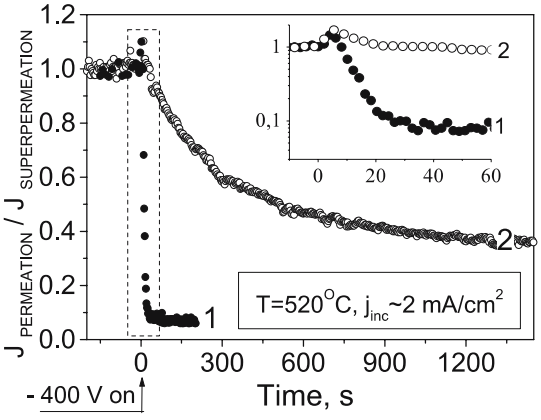


FIGURE 7 Permeation flux drop with time after applying a bias to the membrane: (1) membrane with a subsurface NbC layer (the sputter-sensitive state), and, (2) without NbC (the sputter-resistant state). Curve 2 corresponds to the case of a low concentration of solute O (see curve 2, Fig. 6). The small maximum that appears immediately after the bias has been applied is due to a small temperature increase caused by ion bombardment

One can see that the decay time is orders-of-magnitude shorter in the case of sputter-sensitive state.

4 Discussion

Although it was found that the transition of the membrane into a sputter-resistant state occurs due to the elimination of the subsurface carbide layer, a question still remains: what specific mechanisms secure the stability of superpermeation under sputtering? It would seem that the upstream O monolayer must be at least damaged (if not completely removed) by ion bombardment, and hence the symmetry of hydrogen release should be shifted in favor of reemission. Another question is: what mechanisms are underlying the effects of carbide? The idea of the explanation suggested in this section can be briefly presented as follows.

According to the experimental data, a near-monolayer O coverage remains at the surface over the whole investigated temperature range ($T > 500\,^{\circ}\mathrm{C}$) despite sputtering in the cases of both a sputter-sensitive and a sputter-resistant state. The monolayer is maintained due to the surface segregation of solute O. Our interpretation of the emergence of superpermeation resistant to sputtering is based on the assumption that the thermal release of absorbed hydrogen governing hydrogen reemission occurs via O-vacancies in an O monolayer (the vacancies serve as the active centers of desorption).

If there is no sputtering, a thermodynamic equilibrium between O-vacancies at the surface and O atoms at the surface and in the bulk is maintained. Therefore the inlet and outlet surfaces have equal concentration of O-vacancies, the membrane exhibiting a symmetry with respect to hydrogen evolution at either side. A superpermeable membrane is resistant to sputtering, if the thermal processes maintaining the above equilibrium proceed with a sufficiently high speed not letting the ion bombardment to appreciably enhance the O-vacancy concentration above the equilibrium value and, thus, to destroy the membrane symmetry.

However, the thermal processes responsible for the O-vacancy equilibrium maintenance slow down dramatically in the presence of a subsurface NbC/Nb_2C layer (due to a slower O diffusion in the carbides), and hence sputtering is giving rise to a radical increase in the concentration of O-vacancies at the inlet surface, and to an ensuing shift of hydrogen evolution balance in favor of the back reemission.

4.1 *Peculiarities of the Group Va metals*

The GroupVa metals (V, Nb and Ta) are most suitable to make superpermeable membranes of [1, 18]. Hydrogen atoms absorbed in a membrane made of one of these metals do not only reach the opposite side of the membrane of any realistic thickness (e.g., larger than 1 mm) and temperature (e.g., 200–1000 °C), but are also crossing the membrane many times before the desorption (the condition of 'strong internal reflection' [19] is satisfied). Thus the steady state permeation flux, j_{perm}, in these metals will be always expressed as $j_{\mathrm{perm}} = j_{\mathrm{inc}}\xi \times A$, where j_{inc} is an incident flux, ξ is the probability of absorption (implantation coefficient), and A is a factor of asymmetry of the release of absorbed hydrogen. Under the experimental conditions of this work, hydrogen reemission by the metal virtually occurs only by way of the usual thermal desorption, even though the membrane is acted upon by energetic hydrogen gas [1, 19], and therefore A is determined by ratio of rate constants of the thermal release of absorbed hydrogen through the up- and downstream surfaces. The detailed balance principle links these rate constants with the probabilities of dissociative absorption of thermal H_2 molecules (sticking coefficients) at the upstream (α_{up}) and downstream (α_{down}) surfaces. Thus [1]:

$$A = \alpha_{\mathrm{up}}/(\alpha_{\mathrm{up}} + \alpha_{\mathrm{down}})\,, \tag{1}$$

and,

$$j_{\mathrm{perm}} = j_{\mathrm{inc}}\xi\,\frac{\alpha_{\mathrm{down}}}{\alpha_{\mathrm{up}} + \alpha_{\mathrm{down}}}\,. \tag{2}$$

In the case of natural surfaces, a membrane typically is nearly symmetric, i.e., $\alpha_{\mathrm{up}} \approx \alpha_{\mathrm{down}} \equiv \alpha$. If the energetic gas does not facilitate the thermal absorption/release at the upstream surface through damaging the nonmetallic monolayer by sputtering, the membrane symmetry ($\alpha_{\mathrm{up}} \approx \alpha_{\mathrm{down}} \equiv \alpha$) is keeping stable. Therefore, we have $j_{\mathrm{perm}} \approx 0.5\, j_{\mathrm{inc}}\xi$ in this case, and we are getting superpermeability to suprathermal hydrogen particles (with taking into account that ξ is comparable to 1 and does not depend on T). Thus sputtering may destroy superpermeability of a Group Va metal only by destroying the initial membrane symmetry (viz. by increasing α_{up}).

In the case of an interaction with thermal molecules, one should take $\xi = \alpha$, and thus the flux of molecule driven permeation, $j_{\mathrm{perm}}^{\mathrm{H_2}}$, through a symmetric membrane is given by:

$$j_{\mathrm{perm}}^{\mathrm{H_2}} = 0.5\, j_{\mathrm{inc}}^{\mathrm{H_2}}\alpha\,, \tag{3}$$

where $j_{\mathrm{inc}}^{\mathrm{H_2}}$ is the incident flux of thermal H_2 molecules. Thus the value of α can be easily found from measuring the molecule driven permeation.

4.2 *The role of* O *vacancies in the* O *monolayer coverage*

4.2.1 Generation of O *vacancies by sputtering.* An O monolayer upon Nb surface generates a barrier for the dissociative absorption, which results in α decreasing by orders of magnitude [9, 10, 16] as compared to the case of a clean surface with $\alpha \approx 0.1$ [16, 20]. Though we had plasma driven permeation drastically reduced under sputtering (e.g., Figs. 1 and 2), it still remained orders-of-magnitude higher than one would expect it by (2), if sputtering did make the upstream surface clean (i.e., as expected at $\alpha_{\rm up} \approx 0.1$). This is an evidence of the fact that an O monolayer was staying at the surface all throughout the sputtering process due to a dynamical maintenance by the surface segregation of solute O. Correspondingly, it was concluded [8, 9] that the underlying cause of the degradation of superpermeation under sputtering is a damage (and not a complete removal) of the upstream O monolayer, viz. the generation of O vacancies presumably serving as additional hydrogen absorption-release centers [8, 9]. The increase of O vacancy concentration leads to an increase of $\alpha_{\rm up}$ (with $\alpha_{\rm down}$ staying unchanged) and, consequently, to a change in the membrane symmetry and to a permeation decrease (see (1) and (2)).

4.2.2 The role of equilibrium O *vacancies.* The solubility and mobility of O in Nb are so great [16] that heating of a Nb sample even to moderate T (e.g., 600–700 °C) results in the dissolution of virgin oxide films [12] and in a virtually uniform oxygen distribution across the sample (in contrast to C staying in the subsurface carbide layer up to ~ 1400 °C because of an extremely low solubility of C in Nb [16]). If there are no perturbing factors (such as sputtering), a thermodynamic equilibrium between the surface and bulk O is established (as it is shown below, the exchange between the surface and bulk O is quick enough for the equilibrium to be rapidly reached even at our lowest T):

$$\frac{\Theta_{\rm OS}^{\rm eq}}{1-\Theta_{\rm OS}^{\rm eq}} = \frac{\Theta_{\rm OB}}{1-\Theta_{\rm OB}} \times \exp\frac{\Delta H_{\rm segr}}{kT}\,, \tag{4}$$

where $\Theta_{\rm OS}$ is O coverage of the surface (with $\Theta_{\rm OS}=1$ assumed to be the maximum coverage obtainable by the segregation of solute O), $\Theta_{\rm OB}$ an atomic fraction of O in the bulk (in the state of solution), $\Delta H_{\rm segr}$ the heat of surface segregation of solute O, and superscript 'eq' marks values referring to the equilibrium between surface and bulk O. Due to a high $\Delta H_{\rm segr}$ [13], $\Theta_{\rm OS}^{\rm eq} \approx 1$ (a monolayer is present at the surface) at all the operational T, even though $\Theta_{\rm OB}$ is typically as low as 0.001–0.01.

According to (4), there is a certain equilibrium concentration of O vacancies in the O monolayer, $\Theta_{\rm VS}^{\rm eq} = 1-\Theta_{\rm OS}^{\rm eq}$. By taking into account $\Theta_{\rm OS}^{\rm eq} \approx 1$ and $\Theta_{\rm OB} \ll 1$, one gets:

$$\Theta_{\rm VS}^{\rm eq} = \frac{1}{\Theta_{\rm OB}} \cdot \exp -\frac{\Delta H_{\rm segr}}{kT}\,. \tag{5}$$

Suppose that it is the equilibrium O vacancies that are making active centers of the thermal absorption-desorption of hydrogen when there is no sputtering (as that has been substantiated for the sputtering-sensitive state [8, 9]). Then, the sticking coefficient, $\alpha_{\rm eq}$, for the dissociative absorption of H_2 molecules through an equilibrium O coverage must be proportional to the equilibrium concentration of active centers, i.e.,

$$\alpha_{\rm eq} = A(T)(\Theta_{\rm VS}^{\rm eq})^n = \frac{B(T)}{\Theta_{\rm OB}^n}\,, \tag{6}$$

where $n = 1, 2, 3\ldots$, depending on whether single, or double, or triple, etc., O vacancies actually govern H_2 absorption-desorption (e.g., double centers are typically needed for the reactions of diatomic molecules at the surface); A and B are factors depending only on T. Then, according to (3),

$$j_{\rm perm}^{\rm H_2} \approx 0.5\, j_{\rm inc}^{\rm H_2}\alpha_{\rm eq} = 0.5\, j_{\rm inc}^{\rm H_2}\frac{B(T)}{\Theta_{\rm OB}^n}\,. \tag{7}$$

Thus, if the above idea about the role of surface O vacancies in the kinetics of hydrogen absorption-desorption is true, the molecule driven permeation, $j_{\rm perm}^{\rm H_2}$, is expected to depend on concentration of solute O. The data presented in Fig. 5 demonstrate such a dependence and can be considered as an evidence in support of the said hypothesis.

4.3 *Why sputtering does not affect the membrane symmetry*

Due to O removal by sputtering, the concentration of O in the lattice, $\Theta_{\rm OB}$, should be gradually decreasing and, generally speaking, may become non-uniform, with $\Theta_{\rm OB} = \Theta_{\rm OB}(x, t)$, where x is a transverse coordinate, and t is time. Take $x = 0$ at the upstream surface and $x = l$ at the downstream surface (l is a membrane thickness). Then $\Theta_{\rm OB}(0, t)$, $\Theta_{\rm OB}(l, t)$, $\Theta_{\rm OS}(0, t)$, and $\Theta_{\rm OS}(l, t)$ denote values of the bulk and surface concentrations of O referring to the upstream and downstream surfaces, respectively, while $\Theta_{\rm VS}(0, t)$ and $\Theta_{\rm VS}(l, t)$ is the same for O vacancy concentrations.

According to the above (Sect. 4.1), superpermeability of a Group Va metal will be conserved as long as the membrane remains symmetric ($\alpha_{\rm up} \approx \alpha_{\rm down}$) in spite of only the upstream side being sputtered. Sticking coefficients of thermal H_2 molecules to a continuously sputtered and to an unperturbed surface may be presented, respectively, as (cf.(6)):

$$\alpha_{\rm up} = A(T)\cdot\Theta_{\rm VS}^n(0,\,t) = A(T)\cdot[\Theta_{\rm VS}^{\rm eq}(0,t) + \Delta\Theta_{\rm VS}^{\rm sp}]^n\,, \tag{8}$$

$$\alpha_{\rm down} = A(T)\cdot[\Theta_{\rm VS}^{\rm eq}(l,t)]^n\,, \tag{9}$$

where $\Delta\Theta_{\rm VS}^{\rm sp}$ is an extra O vacancy concentration due to sputtering, while $\Theta_{\rm VS}^{\rm eq}(0, t)$ and $\Theta_{\rm VS}^{\rm eq}(l, t)$ are equilibrium O vacancy concentrations at the up- and downstream surfaces, respectively. According to (8) and (9), the membrane remains symmetric ($\alpha_{\rm up} \approx \alpha_{\rm down}$), as long as, first,

$$\Delta\Theta_{\rm VS}^{\rm sp} \ll \Theta_{\rm VS}^{\rm eq}(0,\,t)\,. \tag{10}$$

(i.e., the concentration of the extra O vacancies generated by sputtering should be negligible compared to the equilibrium value), and, second, if $\Theta_{\rm VS}^{\rm eq}(0, t) \approx \Theta_{\rm VS}^{\rm eq}(l, t)$. According to (5), the latter is valid, as long as:

$$\Theta_{\rm OB}(0,\,t) \approx \Theta_{\rm OB}(l,\,t)\,; \tag{11}$$

(i.e., a removal of O from the upstream side should result in no noticeable non-uniformity in the O distribution over the membrane bulk).

The condition of a uniform distribution of oxygen across the membrane (11) means that, on average, O atoms are crossing the membrane repeatedly before being sputtered, i.e., it should be $\tau_{OB} \gg l^2/6D_O$, where τ_{OB} is the average time that an O atom is spending in the membrane, D_O is diffusion coefficient of O in Nb ($l^2/6D_O$ is the average time of a single crossing of the membrane). If O atoms are crossing the membrane repeatedly before being sputtered, τ_{OB} will be expressed as:

$$\tau_{OB} \approx \frac{N_{me}\Theta_{OB}l}{j_{inc}\gamma\,\Theta_{OS}} \approx \frac{N_{me}\Theta_{OB}l}{j_{inc}\gamma}\,, \tag{12}$$

where N_{me} is the concentration of metallic atoms, j_{inc} an incident ion flux density, and γ, O sputtering coefficient. We took $\Theta_{OS} \approx 1$ in (12) in accordance with the experimental data (e.g., with the fact that superpermeation remains stable under sputtering); a theoretical justification of that is given below. On substituting into the inequality $\tau_{OB} \gg l^2/6D_O$ the (12) and $D_O = D_O^* \times \exp(-E_{OD}/kT)$, where D_O^* and E_{OD} are the pre-exponential factor and activation energy of O diffusion in Nb, respectively, one is getting a condition for $\Theta_{OS}(0, t) \approx \Theta_{OS}(l, t)$ being valid:

$$G \equiv \frac{6N_{me}\Theta_{OB}D_O^*}{j_{inc}\gamma l}\exp-\frac{E_{OD}}{kT} \gg 1\,. \tag{13}$$

Condition (10) is fulfilled, when the rate of the generation of O vacancies by way of sputtering of the surface O atoms is much lower than that by way of O atom transfer from the surface into the lattice (the mechanism which controls the generation of equilibrium O vacancies):

$$j_{inc}\gamma\Theta_{OS} \ll N_S\Theta_{OS}\nu \times \exp-\frac{E_{SB}}{kT}\,. \tag{14}$$

Here N_S is concentration of the sites at the surface that can be occupied by O atoms, ν is an attempt frequency, and E_{SB} is an activation energy for the transition of an O atom from surface into Nb bulk. If there is no specific subsurface layer (e.g., of NbC) that might impede such a transition, it appears reasonable to take $E_{SB} = \Delta H_{serg} + E_{OD}$. With accounting for that, (14) can be expressed as:

$$F \equiv \frac{N_S\nu}{j_{inc}\gamma}\exp-\frac{E_{OD}+\Delta H_{segr}}{kT} \gg 1\,. \tag{15}$$

Presented in Fig. 8 are $G(T)$ and $F(T)$ calculated with $N_{me} = 5\times10^{22}\,\mathrm{cm}^{-3}$, $N_S = 10^{15}\,\mathrm{cm}^{-2}$, $D_O^* = 0.015\,\mathrm{cm}^2/\mathrm{s}$, $E_{OD} = 115\,\mathrm{kJ/mol}$ [16], and $\nu \approx 10^{13}\,\mathrm{s}^{-1}$ for two values of ΔH_{segr} : 42 kJ/mol [21] and 71 kJ/mol [13], and for our experimental conditions ($j_{inc}\gamma \approx 1\times10^{14}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$ and $l = 0.01$ cm). Allowing for a significant uncertainty about the magnitudes of γ, D_O^*, E_{OD} and ΔH_{segr}, the presented graph is not a rigorous estimate; still it shows that $G \gg 1$, and hence $\Theta_{OS}(0, t) \approx \Theta_{OS}(l, t)$, as well as $F \gg 1$, and hence $\Delta\Theta_{VS}^{sp} \ll \Theta_{VS}^{eq}(0, t)$, actually may take place in the operational ranges of T and of other parameters of our experiment.

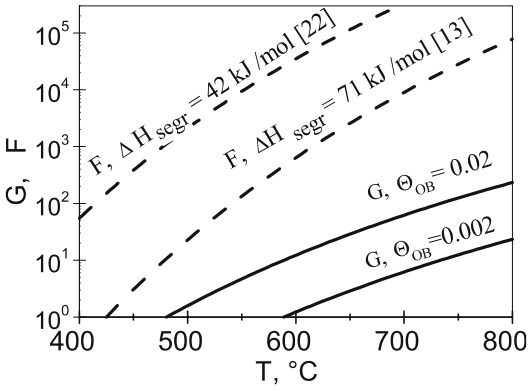


FIGURE 8 Temperature dependencies of G (13) at two values of the initial concentration of solute O, and of F (15) at two values of the heat of surface segregation of O in Nb. The initial membrane symmetry is not noticeably affected by sputtering, when F and G exceed 1. The estimates are made for a uniform (with no NbC layer) membrane sample

Thus sputtering may not be affecting the initial membrane symmetry over a wide range of E_{inc}, j_{inc}, T, Θ_{OB}, and, consequently, the superpermeation resistant to sputtering appears to be a normal property of Nb with a natural or specially introduced bulk O impurity.

Of course, a continuing sputtering, if even it does not perturb the membrane symmetry, will gradually exhaust the bulk O store. Still that is a long-term process, and, besides, the membrane can be doped with O in situ, if necessary [9, 10].

4.4 *Correlation between the resistance to sputtering and the amount of solute* O

According to (12), the higher the O concentration, Θ_{OB}, the greater the average time that an O atom spends in the membrane sample before being taken out by sputtering. This implication may appear rather surprising on the face of it; such a dependence is conditioned by a rather peculiar kinetics of O removal by sputtering: the sputtering rate remains constant at any Θ_{OB} (a zero-order process in Θ_{OB}), as long as we have $\Theta_{OS} \approx 1$ at the surface subject to sputtering($j_{inc}\gamma\Theta_{OS} \approx j_{inc}\gamma$). In the case of equilibrium (4), $\Theta_{OS} \approx 1$ is actually taking place over a wide range of Θ_{OB}, including very low Θ_{OB}, and that also keeps under sputtering, since sputtering does not virtually perturb an equilibrium O coverage, as it is shown in Sect. 4.3. Consequently, the condition (11) of a uniform O distribution across the membrane is fulfilled the easier, the greater Θ_{OB} (see (13)), and hence an addition of O should result in a higher resistance to sputtering.

Corroborating the above notion are the data presented in Fig. 6: superpermeation exists at all the investigated E_{inc} in the case of higher Θ_{OB} (graph1), but at a very small Θ_{OB} (graph2), sputtering still affects the uniform O distribution, resulting in $\Theta_{OB}(0, t) < \Theta_{OB}(l, t)$; that is the cause of superpermeation getting destroyed at higher E_{inc} (i.e., at higher γ).

4.5 *Why does the carbide layer give rise to a sputter-sensitive state*

The permeation is dropping very quickly after the initiation of sputtering in the sputter-sensitive state, i.e., in the

presence of NbC layer (Fig. 7, curve 1); the change of the state of the surface that conditions the permeation decrease must occur, apparently, yet much quicker. That provides a clue for the understanding of the role played by NbC layer: sputtering in the presence of NbC results in a fast permeation drop because it increases the concentration of surface O vacancies (to make $\Delta\Theta_{VS}^{sp} \gg \Theta_{VS}^{eq}$) with virtually no changes in the concentration of solute O and in the O distribution across the membrane (i.e., sputtering distorts the bulk-surface equilibrium (10)). Such an effect takes place due to an impeded O exchange between the surface and the lattice, most probably, because of a radical deceleration of O diffusion in NbC as compared to pure Nb metal [22] (that corresponds to a higher diffusion barrier in the exponent in (15)). An appreciable vacancy concentration increase, $\Delta\Theta_{VS}^{sp}$, can be reached very quickly: e.g., the time needed to sputter off even a full O monolayer (estimated as $\sim 10^{15}/j_{inc}\gamma$) is as short as just a few seconds for our conditions. Since $\Theta_{OB}(0) \approx \Theta_{OB}(l)$ remains valid for the homogeneous region inside the membrane not affected by the presence of the near-surface layer of NbC, permeation is reaching a new stable level after a quick initial drop Fig. 7 [8–10].

The permeation is dropping orders of magnitude slower in the case of sputter-resistant state (Fig. 7, curve 2) when such a drop still occurs at the lowest concentration of solute O (see Fig. 6, curve 2). According to the above estimates (Sect. 4.3), one would expect the equilibrium between the surface and bulk O (condition (10)) not to be noticeably perturbed in the absence of NbC layer, but the condition of a uniform O distribution across the membrane (11) will be violated at a low concentration of solute O. However, a significant portion of solute O has to be removed in this case to produce the effect, and that needs a much longer time, which corresponds to curve 2 in Fig. 7.

4.6 *Why does carbide layer remain stable under the ion bombardment*

One cannot leave out another question: why does sputtering or heating not remove the carbide layer from the upstream side (like it removes the oxygen/oxide layer), with an ensuing membrane transition into a sputter-resistant state? The answer is: Unlike in the case of O, there is no surface segregation of C on Nb, and C solubility in Nb is negligible [16] (less than 0.1 at. % at $T < 1400\,^{\circ}C$). That must occur so due to a particularly high binding energy of C in NbC [16], with NbC phase being thermodynamically by far more preferable than any other state of C in Nb. As a result: (1) C cannot be sputtered as O, i.e., through sputtering off of the top surface layer permanently replenished by segregation from the bulk, (2) C stays localized in the subsurface NbC layer up to rather high temperatures.

Taking that into account that the sputtering of a NbC layer must be determined by the relation between E_{inc} and the energy thresholds of C and Nb sputtering from NbC, E_{th}^{C} and E_{th}^{Nb}, respectively. If $E_{th}^{C} < E_{inc} < E_{th}^{Nb}$, sputtering will stop after the removal of C from the top of NbC layer. Note that $E_{th}^{Nb} \approx 300\,eV/proton$ [23], i.e., ~ 600 and 900 eV for H_2^+ and H_3^+ ions that make the main ion components in our plasma [8, 16]. Thus NbC layer cannot be ever removed at $E_{inc} < 300\,eV/proton$, and that explains why we never saw sputtering remove NbC in our usual range of bias voltages 0–250 V.

If $E_{inc} > E_{th}^{Nb}$, sputtering must gradually remove the NbC layer, but with a low rate characteristic of the sputtering of the heavy Nb atoms by light hydrogen ions. Still the upstream carbide layer will be sputtered away in the end in this case. As a result, the long-term evolution of permeation is expected to be non-monotonic. First, an initiation of sputtering is to cause a quick degradation of superpermeation, with the permeation flux dropping by orders of magnitude because of the presence of the NbC layer impeding the surface segregation of solute O. But in a longer run, a complete recovery of superpermeation is expected to occur upon the complete NbC layer removal. Certainly, that only can be possible, if a sufficient amount of O remains in the sample after the long-term sputtering, but the characteristic time of O removal can be controlled by varying the initial O concentration (see (12)). Besides, an additional amount of solute O can be added up in situ, if necessary.

5 Conclusions

In contrast to the usual sensitivity of superpermeation to sputtering, a stable superpermeation of hydrogen in Nb was observed over the whole investigated range of incident ion energies (up to 1 keV) and metal temperatures ($T \geq 520\,^{\circ}C$). Superpermeability resistant to sputtering appears to be a common property of Nb (and most probably, of V and Ta as well) containing a natural or specially introduced O impurity.

A subsurface carbide layer usually present at a 'real'/'natural' surface of carbide-forming metals is found to be responsible for the formerly observed sensitivity of superpermeation to sputtering because such a layer impedes the exchange between the surface and bulk O.

Doping of the metal with an extra amount of O is found to be a good means of additionally enhancing the resistance of decarbidized superpermeable Nb membrane to sputtering.

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